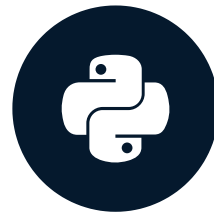


What are multi-agent systems?

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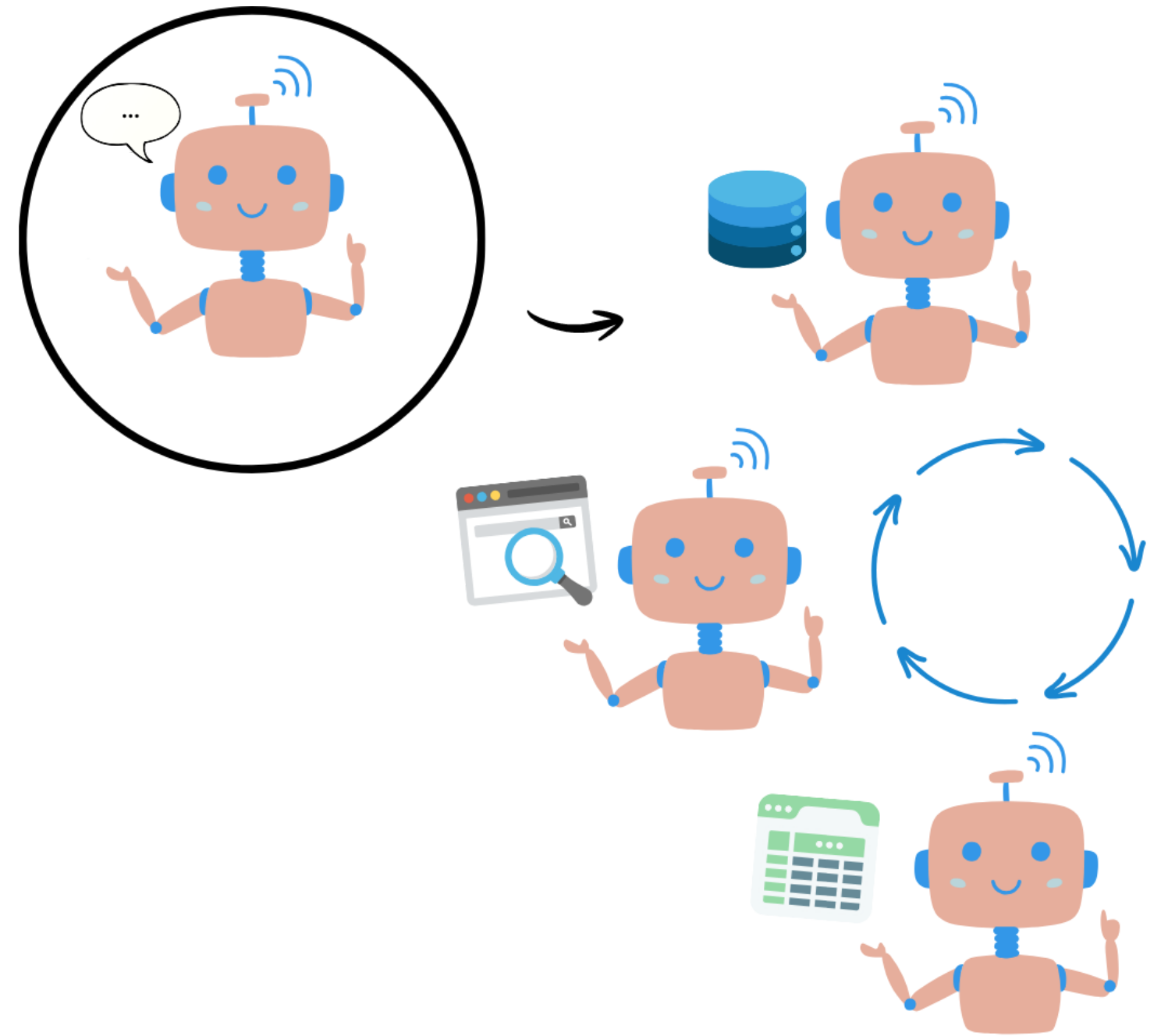


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Beyond one agent

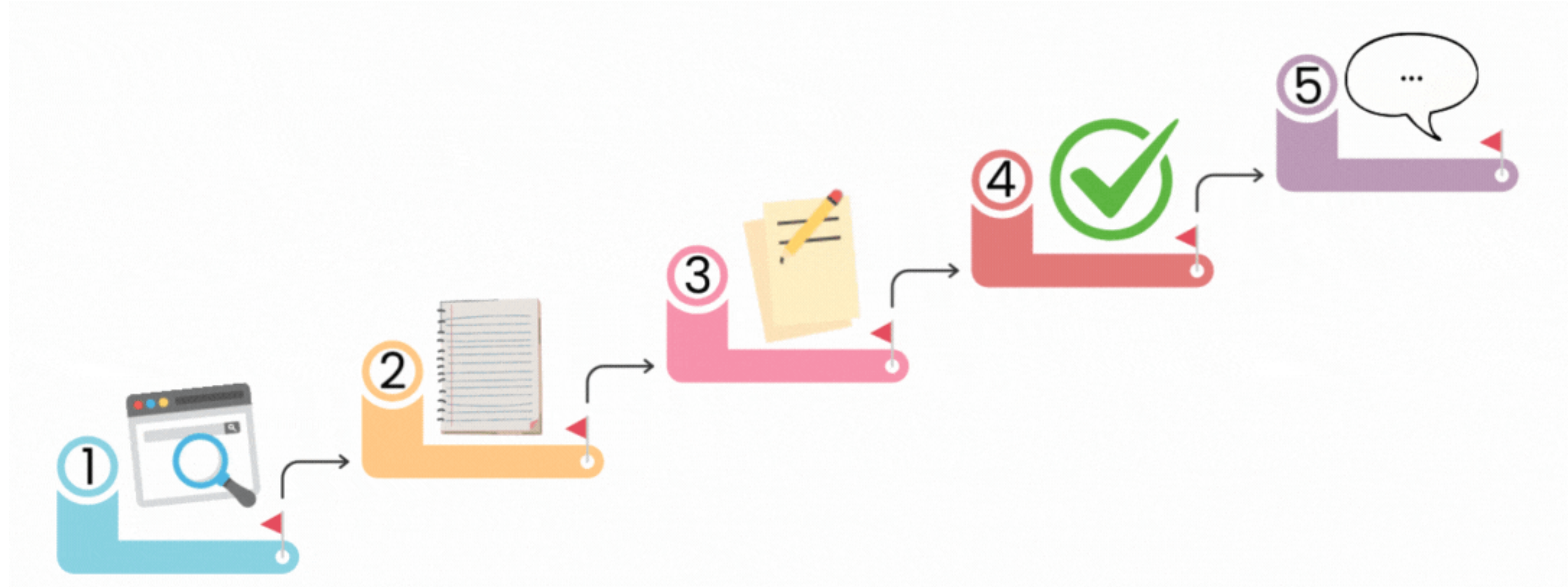
- **Single agent:** powerful, but limited
- **Specialized agents** can be more effective
- Can collaborate as a **team of agents**



Why use many agents?

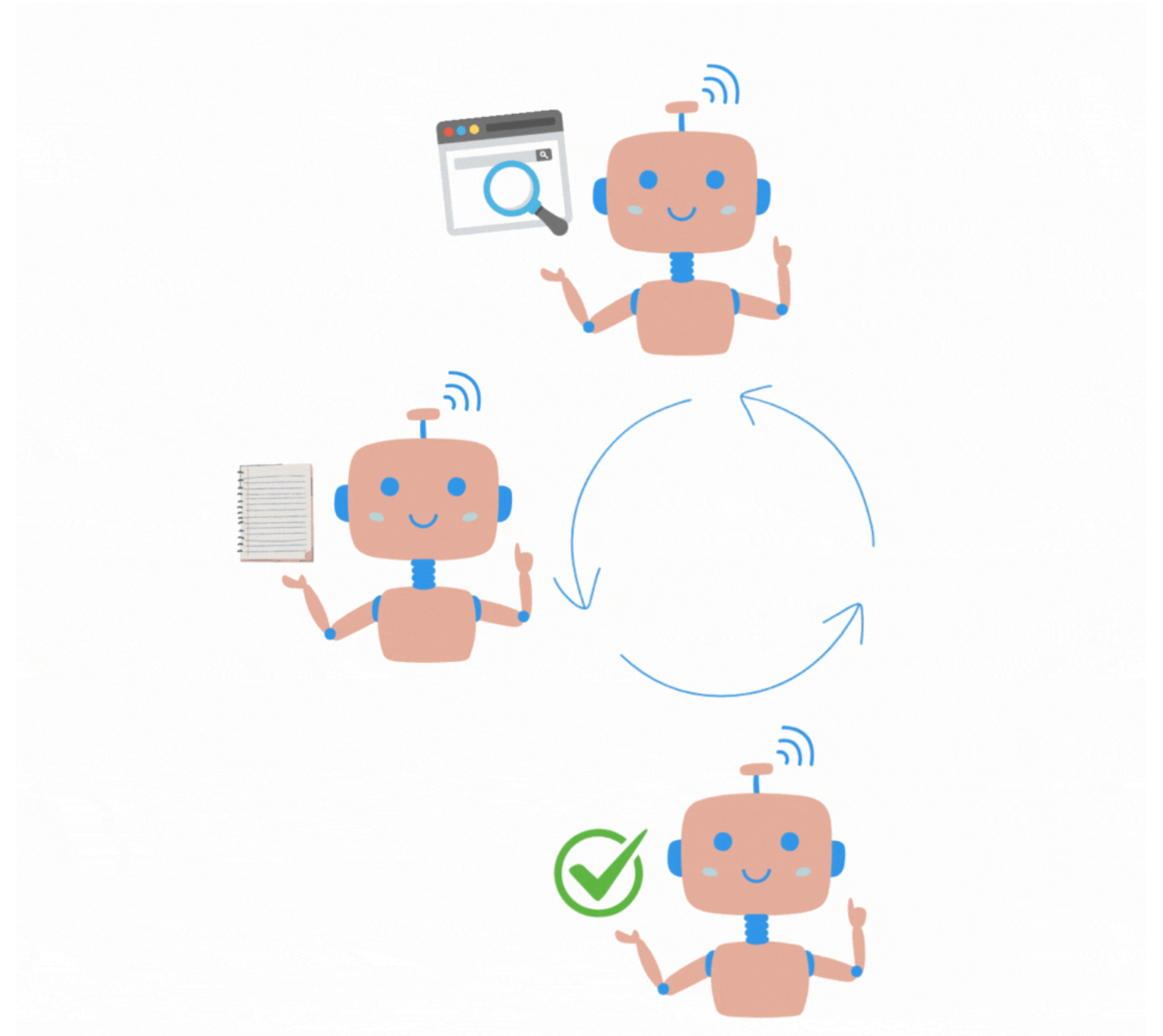
Example:

- Writing a research report
- One agent juggling: searching, compiling, writing, checking, and returning the answer



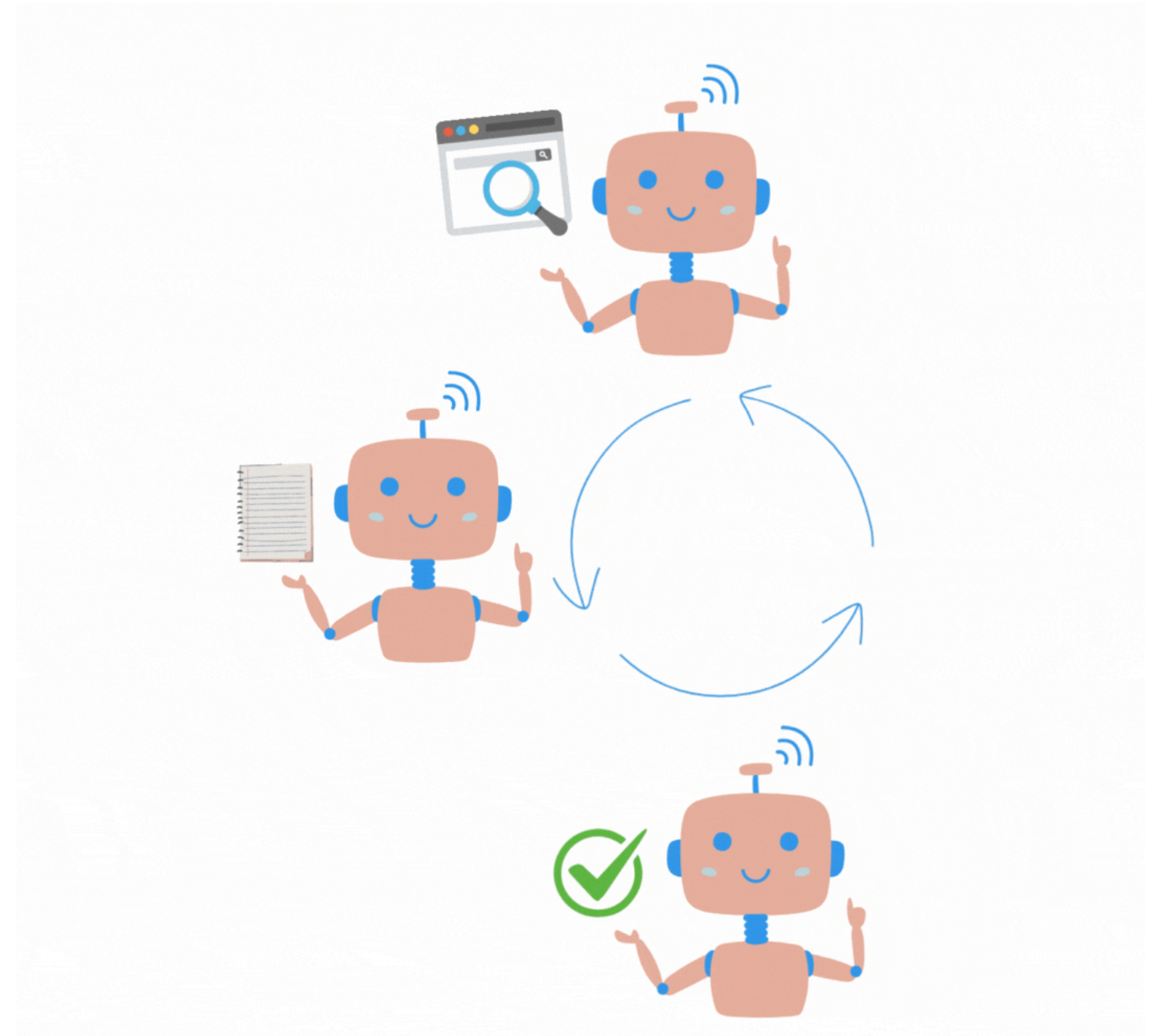
Meet the specialized agents

- **Research Agent:** gathers information
- **Write Agent:** drafts the report
- **Review Agent:** checks and improves content



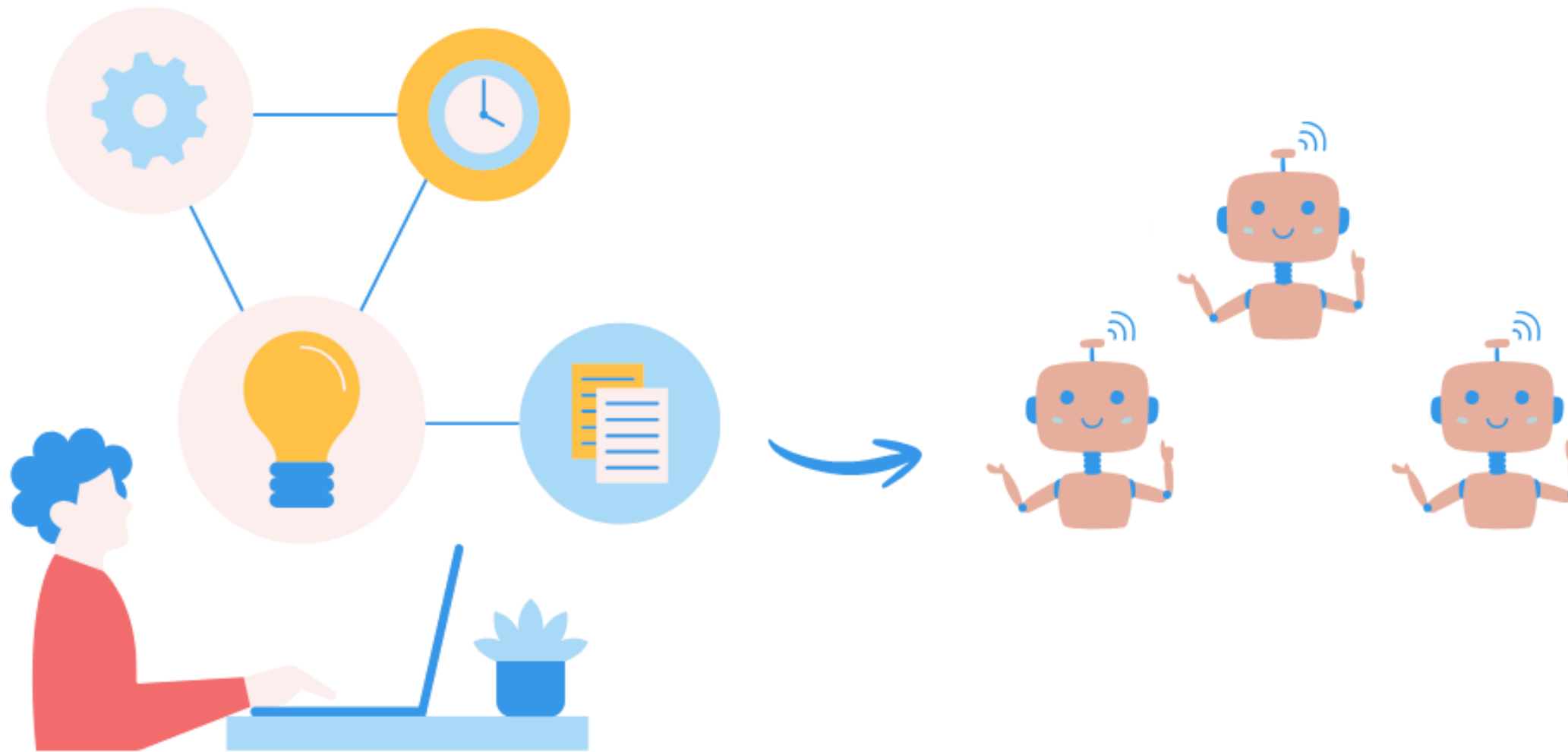
How they work together

- **Research Agent** collects notes
- **Write Agent** composes draft using notes
- **Review Agent** refines final output



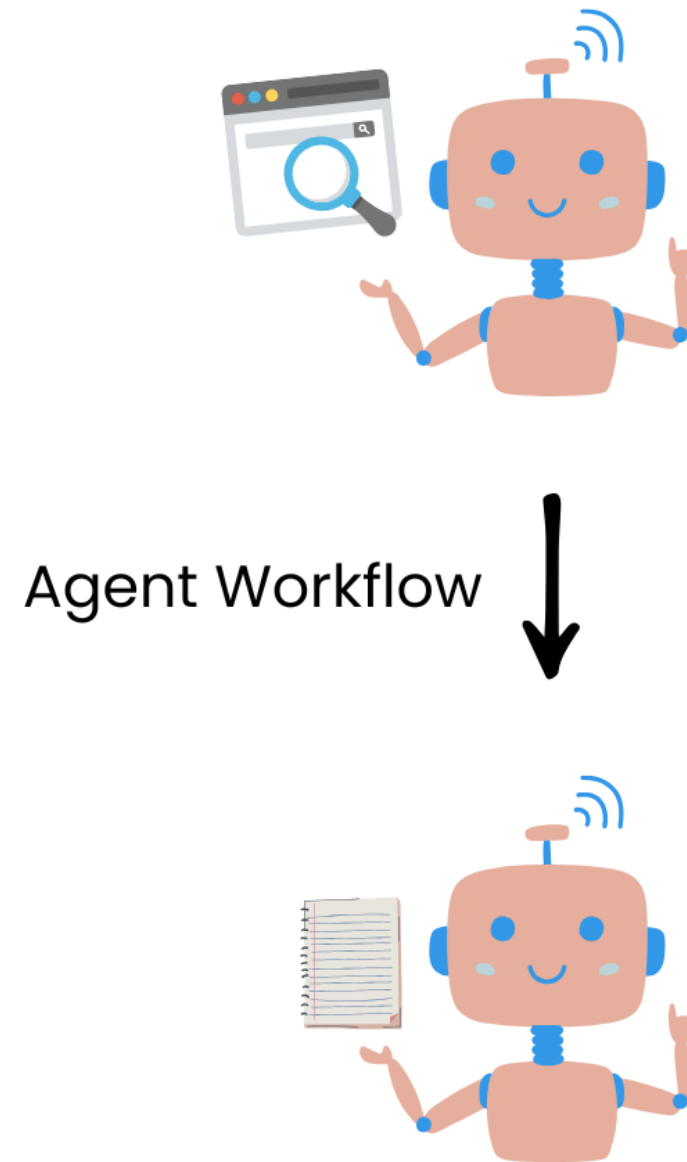
When to scale up

- If tasks naturally split into parts
- If one agent struggles with complex prompts or too many tools



Built-in AgentWorkflow

- LlamaIndex AgentWorkflow
- Supports handing tasks between agents automatically
- Brings a layer of complexity

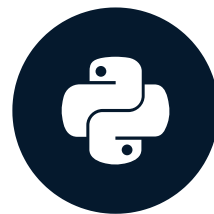


Let's practice!

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Multi-agent system with AgentWorkflow

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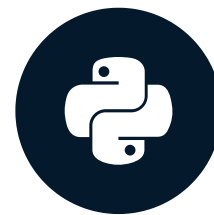
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Let's practice!

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Building agentic workflows from scratch

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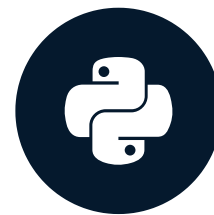
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Let's practice!

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Designing multi-step workflows with custom events

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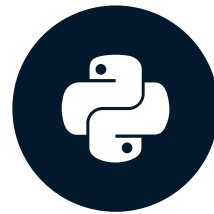
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Let's practice!

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Concurrency and event collection

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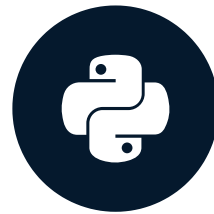
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Let's practice!

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Putting together a multi-agent system

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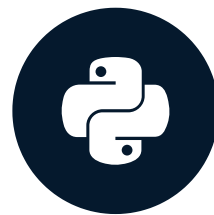
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Adding self- reflection to the workflow

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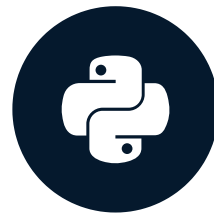
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Let's practice!

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Wrap-up

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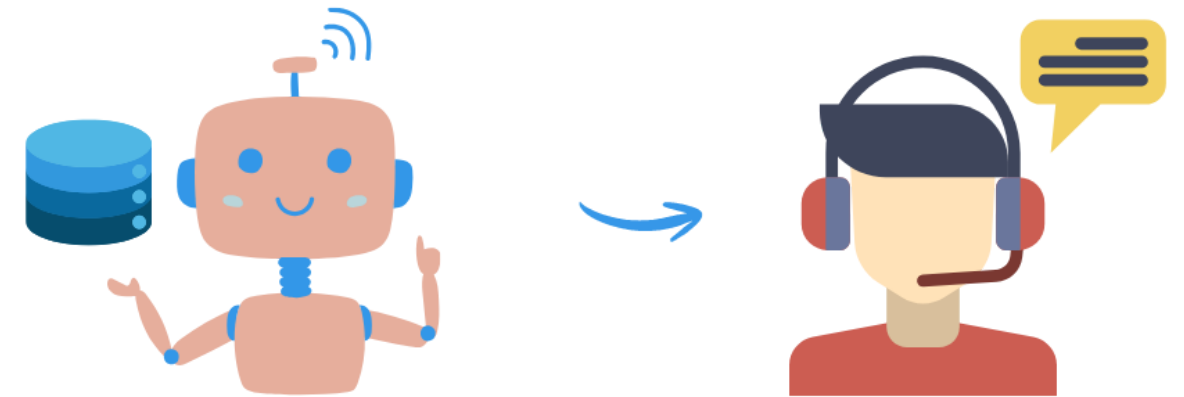
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What you can build

Some AI project ideas:

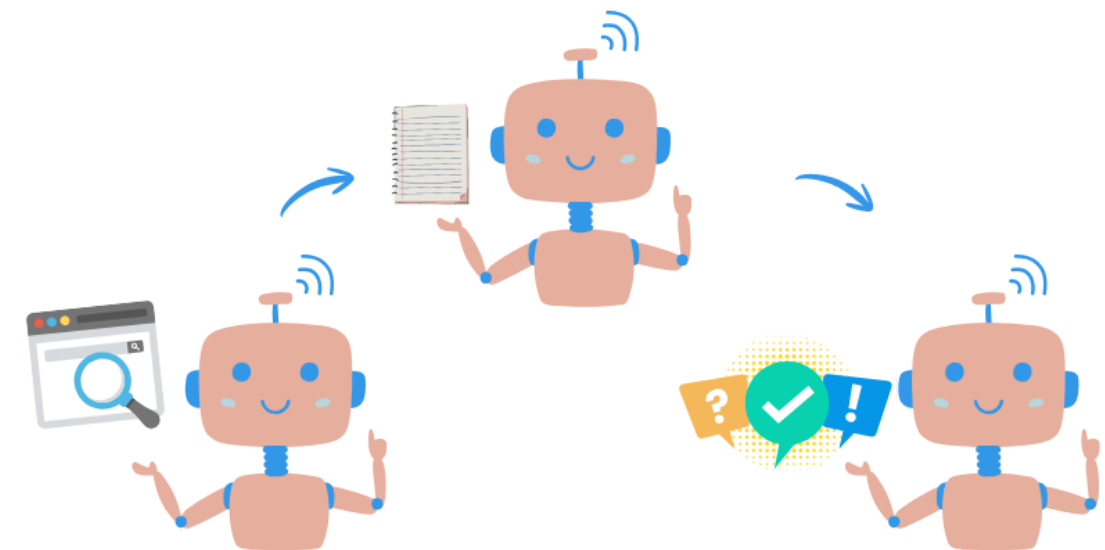
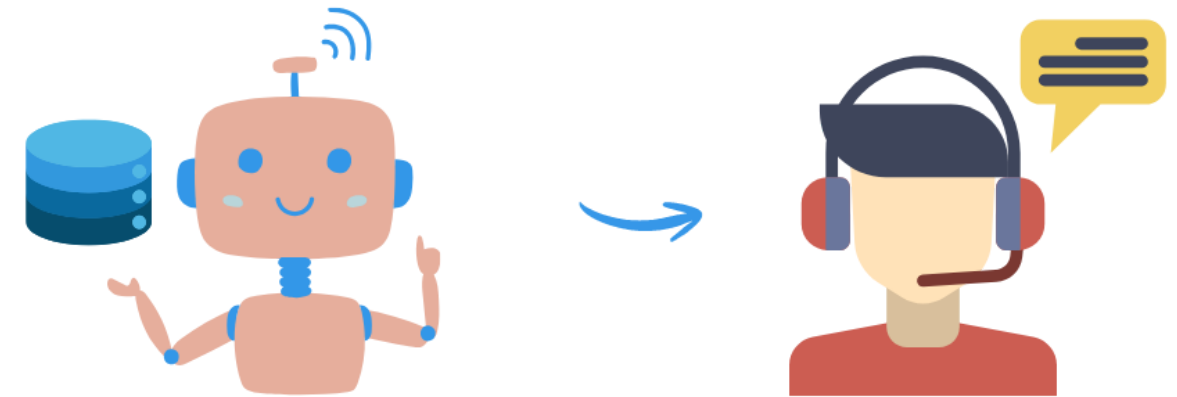
- Customer support bot with knowledge base lookup



What you can build

Some AI project ideas:

- Customer support bot with knowledge base lookup
- AI that researches, writes, then generates quizzes



What's next

- New techniques are emerging rapidly:
 - Such as the self-reflection loops we've seen
- Explore [LlamaIndex](#) latest updates and docs
- Practice by building projects in DataLab



LlamaIndex



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Congratulations!

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